

$D=4$, Lorentz: $SO(1,3)$

Poincare: $SO(1,3) \ltimes \mathbb{R}^{1,3}$

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu} \quad \text{and} \quad x'^2 = x^2 = \eta_{\mu\nu} x^{\mu} x^{\nu}$$

$$\Rightarrow \Lambda^{\mu}_{\sigma} x^{\sigma} \Lambda^{\nu}_{\lambda} x^{\lambda} \eta_{\mu\nu} = x^{\sigma} x^{\lambda} \eta_{\sigma\lambda}$$

$$\text{or } \Lambda^{\mu}_{\sigma} \Lambda^{\nu}_{\lambda} \eta_{\mu\nu} = \eta_{\sigma\lambda}$$

$\rightarrow$ Inf. near $\mathbb{1}$:

$$\Lambda^{\mu}_{\sigma} = \delta^{\mu}_{\sigma} + \omega^{\mu}_{\nu} + \mathcal{O}(\omega^2)$$

$\odot \Lambda^{\mu}_{\nu}$ give a rep of the Lorentz group

$$\mathcal{U}: G \rightarrow \text{Aut}(V)$$

$$\mathcal{U}: SO(1,3) \rightarrow \text{Aut}(V)$$

$$\mathcal{U}(\Lambda): V \rightarrow V$$

$$\text{Here } V = \mathbb{R}^{1,3}$$

"Self rep"

$$\odot \mathcal{U}(\Lambda) \mathcal{U}(\Lambda') = \mathcal{U}(\Lambda \Lambda')$$

$$\rightarrow \mathcal{U}(\mathbb{1} + \omega)$$

$$= \mathcal{U}(\mathbb{1}) + \omega_{\mu\nu} \mathcal{U}(\omega)$$

$$= \mathbb{1} + \frac{i}{2} \omega_{\mu\nu} M^{\mu\nu}$$

$$M^{\mu\nu} = -M^{\nu\mu} \text{ Acts on } \mathcal{H} \text{ in QM}$$

$\rightarrow M$: Generator

* Lorentz Algebra defined by $\mathcal{U}(\Lambda)^{-1} \mathcal{U}(\Lambda') \mathcal{U}(\Lambda) = \mathcal{U}(\Lambda^{-1} \Lambda' \Lambda) f$

$$f \mapsto \Lambda f$$

$$f \xrightarrow{\Lambda} \Lambda f$$

$$\downarrow \Lambda'$$

$$\Lambda' \Lambda f$$

* Idea: does $\Lambda^{-1} \Lambda' \Lambda$

close and if not

this tells us $\Lambda^{-1} \Lambda' \Lambda f \leftarrow \Lambda^{-1}$

how much the contents of each group el.

now when we take the limit of each group as
moves us away from its "unbalancedness"

$$* \mathcal{U}(\Lambda)^{-1} \mathcal{U}(\Lambda') \mathcal{U}(\Lambda) = \mathcal{U}(\Lambda^{-1} \Lambda' \Lambda)$$

expand to linear order in ω^a

$$\begin{aligned} \Lambda' &= \mathbb{1} + \omega \\ \mathcal{U}(\Lambda^{-1})^{-1} \left(\mathbb{1} + \frac{i}{2} \omega_{\mu\nu} M^{\mu\nu} \right) \mathcal{U}(\Lambda) & \quad (\text{LHS}) \\ &= \mathcal{U}(\Lambda^{-1} (\mathbb{1} + \omega \Lambda)) \\ &= \mathcal{U}(\mathbb{1} + \Lambda^{-1} \omega \Lambda) \\ &= \mathbb{1} + \frac{i}{2} \Lambda_{\mu}^{\sigma} \omega_{\sigma\lambda} \Lambda_{\nu}^{\lambda} M^{\mu\nu} \quad (\text{RHS}) \end{aligned}$$

$$\boxed{\mathcal{U}(\Lambda)^{-1} M^{\mu\nu} \mathcal{U}(\Lambda) = \Lambda_{\rho}^{\mu} \Lambda_{\kappa}^{\nu} M^{\rho\kappa}} \quad (\text{i})$$

$$* \text{In general: } \mathcal{U}(\Lambda)^{-1} P^{\mu} \mathcal{U}(\Lambda) = \Lambda_{\sigma}^{\mu} P^{\sigma} \quad (\text{ic})$$

* Now expand $\Lambda = \mathbb{1} + \omega$

$$\begin{aligned} (\text{i}) \quad & \left(\mathbb{1} - \frac{i}{2} \omega_{\sigma\lambda} M^{\sigma\lambda} \right) (M^{\mu\nu}) \left(\mathbb{1} + \frac{i}{2} \omega_{\alpha\beta} M^{\alpha\beta} \right) \\ &= \left(\delta_{\rho}^{\mu} + \omega_{\rho}^{\mu} \right) \left(\delta_{\kappa}^{\nu} + \omega_{\kappa}^{\nu} \right) M^{\rho\kappa} \end{aligned}$$

$$\begin{aligned} & \left(M^{\mu\nu} - \frac{i}{2} \omega_{\sigma\lambda} M^{\sigma\lambda} M^{\mu\nu} \right) \left(\mathbb{1} + \frac{i}{2} \omega_{\alpha\beta} M^{\alpha\beta} \right) \\ &= \left(\delta_{\rho}^{\mu} \delta_{\kappa}^{\nu} + \omega_{\rho}^{\mu} \delta_{\kappa}^{\nu} + \delta_{\rho}^{\mu} \omega_{\kappa}^{\nu} \right) M^{\rho\kappa} \end{aligned}$$

$$= M^{\mu\nu} + \omega_{\sigma}^{\mu} M^{\sigma\nu} + M^{\mu\sigma} \omega_{\sigma}^{\nu}$$

$$\frac{1}{2} \left(M^{\mu\nu} M^{\sigma\lambda} - M^{\sigma\lambda} M^{\mu\nu} \right) \omega_{\sigma\lambda}$$

$$\hookrightarrow \omega_{\sigma\lambda} = \eta^{\mu\lambda} \left(\omega_{\sigma\lambda} M^{\sigma\nu} + M^{\mu\sigma} \omega_{\sigma\lambda} \eta^{\lambda\nu} \right)$$

$$\begin{aligned} \omega_{\sigma\lambda} M^{\sigma\nu} &= \frac{1}{2} \omega_{\sigma\lambda} (M^{\sigma\nu} - M^{\nu\sigma}) \\ \text{and } M^{\mu\sigma} \omega_{\sigma\lambda} &= \frac{1}{2} \omega_{\sigma\lambda} (M^{\mu\sigma} - M^{\sigma\mu}) \end{aligned}$$

$$\omega_{\sigma\lambda} M^{\sigma\nu} = \frac{1}{2} (\omega_{\sigma\lambda} M^{\sigma\nu} - \omega_{\sigma\lambda} M^{\nu\sigma})$$

$$\frac{1}{2} [M^{\mu\nu}, M^{\mu\nu}] = 2 (M^{\mu\nu} - M^{\nu\mu}) / \hbar + \hbar (M^{\mu\nu} - M^{\nu\mu})$$

$$[M^{\mu\nu}, M^{\rho\kappa}] = i (\eta^{\mu\kappa} M^{\nu\rho} + \eta^{\kappa\nu} M^{\rho\mu} - \eta^{\kappa\mu} M^{\rho\nu} - \eta^{\mu\nu} M^{\rho\kappa})$$

Restart:

$$\psi(\Lambda)^{-1} M^{\mu\nu} \psi(\Lambda) = \Lambda^\mu_\rho \Lambda^\nu_\kappa M^{\rho\kappa}$$

$$(1 - \frac{i}{2} \omega_{\rho\kappa} M^{\rho\kappa}) M^{\mu\nu} (1 + \frac{i}{2} \omega_{\rho\kappa} M^{\rho\kappa})$$

$$\Rightarrow M^{\mu\nu} + \frac{i}{2} \omega_{\rho\kappa} [M^{\mu\nu}, M^{\rho\kappa}]$$

$$= (\delta^\mu_\rho + \omega^\mu_\rho) (\delta^\nu_\kappa + \omega^\nu_\kappa) M^{\rho\kappa}$$

$$= (\delta^\mu_\rho \delta^\nu_\kappa + \omega^\mu_\rho \delta^\nu_\kappa + \delta^\mu_\rho \omega^\nu_\kappa + \omega^\mu_\rho \omega^\nu_\kappa) M^{\rho\kappa}$$

$$= M^{\mu\nu} + \omega^\mu_\rho M^{\rho\nu} + M^{\mu\kappa} \omega^\nu_\kappa$$

$$= \eta^{\mu\kappa} \omega_{\kappa\rho} M^{\rho\nu} + \eta^{\nu\rho} M^{\mu\kappa} \omega_{\kappa\rho}$$

$$= \frac{1}{2} \eta^{\mu\kappa} (\omega_{\mu\rho} - \omega_{\rho\mu}) M^{\rho\nu}$$

$$= \frac{1}{2} \eta^{\mu\kappa} \omega_{\kappa\rho} M^{\rho\nu} - \frac{1}{2} \eta^{\mu\kappa} \omega_{\rho\mu} M^{\rho\nu}$$

$$= \frac{1}{2} \eta^{\mu\kappa} \omega_{\mu\rho} M^{\rho\nu} - \frac{1}{2} \eta^{\mu\rho} \omega_{\kappa\rho} M^{\mu\nu}$$

$$= \omega_{\kappa\rho} (\frac{1}{2}) (\eta^{\mu\kappa} M^{\rho\nu} - \eta^{\mu\rho} M^{\kappa\nu})$$

$$\therefore (RHS) = \frac{1}{2} \omega_{\kappa\rho} \left[\begin{array}{l} -\eta^{\mu\kappa} M^{\rho\nu} + \eta^{\mu\rho} M^{\kappa\nu} \\ \eta^{\nu\rho} M^{\mu\kappa} = \eta^{\nu\kappa} M^{\mu\rho} \end{array} \right]$$

$$[M^{\mu\nu}, M^{\rho\kappa}] = i [\eta^{\mu\kappa} M^{\nu\rho} - \eta^{\mu\rho} M^{\kappa\nu} - \eta^{\nu\rho} M^{\mu\kappa} + \eta^{\nu\kappa} M^{\mu\rho}]$$

okay, off by a minus sign...

Lorentz alg. of the momentum operator

$$\psi(p) P^\mu \psi(\Lambda) = \Lambda^\mu_\rho P^\rho$$

$$\begin{aligned}
 & \left(1 - \frac{i}{2} \omega_{\mu\nu} M^{\mu\nu} \right) P^\rho \left(1 + \frac{i}{2} \omega_{\mu\nu} M^{\mu\nu} \right) \\
 & P^\rho + \frac{i}{2} \omega_{\mu\nu} [P^\rho, M^{\mu\nu}] \\
 & = (\delta^\rho_\mu + \omega^\rho_\nu P^\nu) \\
 & = P^\rho + \eta^{\rho\mu} \omega_{\mu\nu} P^\nu \\
 & \therefore \frac{i}{2} [P^\rho, M^{\mu\nu}] = \frac{1}{2} \eta^{\rho\mu} (\omega_{\mu\nu} - \omega_{\nu\mu}) P^\nu \\
 & = \frac{1}{2} (\eta^{\rho\mu} P^\nu - \eta^{\rho\nu} P^\mu) \omega_{\mu\nu}
 \end{aligned}$$

$$[M^{\mu\nu}, P^\rho] = i (\eta^{\rho\mu} P^\nu - \eta^{\rho\nu} P^\mu)$$

off by -1 again...
 * generated by angular momentum & spin
 $(M^{\mu\nu})^i_j = i \left(x^\mu \frac{\partial}{\partial x^\nu} - x^\nu \frac{\partial}{\partial x^\mu} \right) \delta^i_j + (S^{\mu\nu})^i_j$

* Spin also obeys $[S, S]$ as $[M, M]$

* Classification of each rep:

Rotations: $J_i \equiv \frac{1}{2} \epsilon_{ijk} M_{jk}$ $ijk = 1, 2, 3$
 Boosts: $K_i \equiv M_{0i}$

↓ last try:

$$\begin{aligned}
 & \mathcal{U}(\Lambda^{-1}) P^\mu \mathcal{U}(\Lambda) = \Lambda^\mu_\sigma P^\sigma \\
 & \left(1 + \frac{i}{2} \omega_{\alpha\beta} M^{\alpha\beta} \right) P^\mu \left(1 + \frac{i}{2} \omega_{\sigma\lambda} M^{\sigma\lambda} \right) = (\delta^\mu_\sigma + \omega^\mu_\sigma) P^\sigma \\
 & \left(P^\mu - \frac{i}{2} \omega_{\alpha\beta} M^{\alpha\beta} P^\mu \right) \left(1 + \frac{i}{2} \omega_{\sigma\lambda} M^{\sigma\lambda} \right) = (\delta^\mu_\sigma + \omega^\mu_\sigma) P^\sigma \\
 & \therefore \left(P^\mu - \frac{i}{2} \omega_{\sigma\lambda} P^\mu M^{\sigma\lambda} + \frac{i}{2} \omega_{\alpha\beta} M^{\alpha\beta} P^\mu \right) = (\delta^\mu_\sigma + \omega^\mu_\sigma) P^\sigma \\
 & \therefore P^\mu + \omega_{\sigma\lambda} \left(\frac{i}{2} \right) (M^{\sigma\lambda} P^\mu - P^\mu M^{\sigma\lambda}) = P^\mu + \omega^\mu_\sigma P^\sigma
 \end{aligned}$$

$$\begin{aligned}
 \omega_{\alpha\lambda} \frac{i}{2} [M^{\sigma\lambda}, p^\mu] &= \omega_{\alpha\sigma} p^\sigma \\
 &= \eta^{\mu\lambda} \omega_{\lambda\sigma} p^\sigma \\
 &= \eta^{\mu\lambda} \left(\frac{1}{2} \right) (\omega_{\lambda\sigma} - \omega_{\sigma\lambda}) p^\sigma \\
 &= \left(\frac{1}{2} \right) (\omega_{\lambda\sigma}) p^\sigma \eta^{\mu\lambda} \\
 &= -\frac{1}{2} \omega_{\lambda\sigma} \eta^{\mu\sigma} p^\lambda
 \end{aligned}$$

$$\therefore [M^{\sigma\lambda}, p^\mu] = -i (p^\sigma \eta^{\mu\lambda} - p^\lambda \eta^{\mu\sigma})$$

Lorentz commutation relation for $SO(1,3) \times \mathbb{R}^{1,4}$

Field theory: 4-comp Dirac Ψ

$$\mathcal{L} = i \bar{\Psi} \gamma^\mu \partial_\mu \Psi - m \bar{\Psi} \Psi \quad \psi = \begin{pmatrix} \psi \\ \bar{\psi} \end{pmatrix}$$

Conventions:

$$\eta_{\alpha\beta} = (-1, 1, 1, 1)$$

$$(\sigma^\mu)_{\alpha\dot{\beta}} = (1, \sigma^i)_{\alpha\dot{\beta}}$$

$$(\bar{\sigma}^\mu)^{\dot{\alpha}\beta} = \epsilon^{\dot{\alpha}\dot{\beta}} \epsilon^{\alpha\beta} (\sigma^\mu)_{\alpha\dot{\beta}}$$

$$\begin{cases} \epsilon^{\alpha\beta} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \epsilon^{\dot{\alpha}\dot{\beta}} \\ \epsilon_{\alpha\beta} = -\epsilon^{\alpha\beta} \end{cases}$$

* Recall:

$$\{Q_\alpha, \bar{Q}_{\dot{\alpha}}\} = (C \sigma^\mu_{\alpha\dot{\alpha}}) P_\mu \quad | \quad \{\gamma^\mu, \gamma^\nu\} = -2\eta^{\mu\nu}$$

$$\gamma^\mu = \begin{pmatrix} 0 & (\sigma^\mu)_{\alpha\dot{\beta}} \\ (\bar{\sigma}^\mu)^{\dot{\alpha}\beta} & 0 \end{pmatrix}$$

$$\bar{\Psi} := \Psi^\dagger \beta \quad \text{where}$$

$$\beta \equiv \begin{pmatrix} 0 & \delta^{\dot{\alpha}\beta} \\ \delta_{\alpha\dot{B}} & 0 \end{pmatrix}$$

The equation of motion is given by:

$$(-i \not{\partial} - m) \Psi = 0$$

* K.G:

$$\begin{aligned}
 & (i \not{\partial} + m)(-i \not{\partial} + m) \Psi \\
 &= (\not{\partial} \not{\partial} + m^2) \Psi
 \end{aligned}$$

claim: $\not{\partial} \not{\partial} = -\partial^2 = -\square^2$

proof: $\not{\partial} \not{\partial} = \gamma^\mu \gamma^\nu \partial_\mu \partial_\nu$

$$= -2\eta^{\mu\nu} \partial_\mu \partial_\nu - \gamma^\nu \gamma^\mu \partial_\mu \partial_\nu$$

• 1st second indices:

$$\partial\partial = -2\eta^{\mu\nu}\partial_\mu\partial_\nu = \gamma^\mu\gamma^\nu\partial_\mu\partial_\nu = \gamma^\mu\gamma^\nu\partial_\nu\partial_\mu$$

$$\therefore \partial\partial = -\eta^{\mu\nu}\partial_\mu\partial_\nu = -\partial^2$$

$$\therefore (-\partial^2 + m^2)\psi = 0$$

solve via F.T.:

$$\mathcal{K}\{(-\partial^2 + m^2)\psi\} = 0$$

$$\Rightarrow (p^2 + m^2)\hat{\psi} = 0$$

requiring on-shell $p^2 = -m^2$

• Plane wave solutions:

$$\psi(x) = u(p)e^{ipx} + v(p)e^{-ipx}$$

Plugging this back into the Dirac Equation

$$(-i\not{\partial} - m)(u(p)e^{ipx} + v(p)e^{-ipx}) = 0$$

$$[-i\gamma^\mu \not{p}_\mu u(p)e^{ipx} - i\gamma^\mu p_\mu (-i)v(p)e^{-ipx}] = 0$$

$$\Rightarrow [\gamma^\mu p_\mu u(p) - m u(p)]e^{ipx} + [-\not{p} v(p) - m v(p)]e^{-ipx} = 0$$

• Since e^{ipx} & e^{-ipx} are linearly ind. their coefficients must each vanish:

$$\therefore \begin{cases} (\not{p} - m)u(p) = 0 \\ (-\not{p} - m)v(p) = 0 \end{cases}$$

* Lorentz invariant measure: $\widetilde{d^3p} \equiv \frac{d^3p}{(2\pi)^3 2E_p}$

$$\psi(x) = \sum_{s=\pm} \int \widetilde{d^3p} (b_s(p)u_s(p)e^{ipx} + d_s^\dagger(p)v_s(p)e^{-ipx})$$

and quantize $(b, d) \rightarrow (b^\dagger, d^\dagger)$ which become anticommuting, forming the u, v components

L_0 be carrying

$u = (u_+, u_-)$ $v = (v_-, v_+)$
define spinor bilinears as follows:

$$[\chi \lambda]_{\pm} = \chi^{\alpha} \lambda_{\alpha} \quad | \quad [\chi \lambda]_{\pm} = \chi_{\dot{\alpha}} \lambda^{\dot{\alpha}}$$

SO(2) SO(2)

and define the product

$$\gamma^{\mu} = \begin{pmatrix} 0 & (\sigma^{\mu})_{\alpha\dot{\beta}} \\ (\bar{\sigma}^{\mu})^{\dot{\beta}\alpha} & 0 \end{pmatrix}$$

$$[\chi | \gamma^{\mu} | \lambda] = \chi^{\alpha} \begin{pmatrix} 0 & (\sigma^{\mu})_{\alpha\dot{\beta}} \\ (\bar{\sigma}^{\mu})^{\dot{\beta}\alpha} & 0 \end{pmatrix} \lambda^{\dot{\beta}}$$

$$= \chi^{\alpha} \sigma^{\mu}_{\alpha\dot{\beta}} \lambda^{\dot{\beta}}$$

Define $\not{P} = P_{\mu} \gamma^{\mu} = \begin{pmatrix} 0 & P_{\mu} (\sigma^{\mu})_{\alpha\dot{\beta}} \\ P_{\mu} (\bar{\sigma}^{\mu})^{\dot{\beta}\alpha} & 0 \end{pmatrix}$

or, rather, $\not{P} = \begin{pmatrix} 0 & P_{\alpha\dot{\beta}} \\ P^{\dot{\alpha}\alpha} & 0 \end{pmatrix}$

* For $\sigma^{\mu} = (1, \sigma^i)$

$$P_{\alpha\dot{\beta}} = P_{\mu} (\sigma^{\mu})_{\alpha\dot{\beta}} = P_0 \sigma^0 + P_i \sigma^i$$

$$= \begin{pmatrix} P_0 & 0 \\ 0 & P_0 \end{pmatrix} + \dots$$

$\Rightarrow \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$P_{\alpha\dot{\beta}} = \begin{pmatrix} P_0 & 0 \\ 0 & P_0 \end{pmatrix} + \begin{pmatrix} 0 & P_1 \\ P_1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & -iP_2 \\ iP_2 & 0 \end{pmatrix} + \begin{pmatrix} P_3 & 0 \\ 0 & -P_3 \end{pmatrix}$$

$$P_{\alpha\dot{\beta}} = \begin{pmatrix} P_0 + P_3 & P_1 - iP_2 \\ P_1 + iP_2 & P_0 - P_3 \end{pmatrix}$$

Since $P_0 = M_{0m} P^m = -P^0$

$P_i = M_{im} P^m = P^i$

$$P_{\alpha\beta} = \begin{pmatrix} -P^0 + P^3 & P^1 - iP^2 \\ P^1 + iP^2 & P^0 - P^3 \end{pmatrix}$$

$\sigma: \mathbb{R}^{1,4} \rightarrow M_{2 \times 2}(\mathbb{C})$

* This isomorphism is Lorentz invariant:

$$\det(P_{\alpha\beta}) = -P^m P_m = m^2$$

* Here we will work with massless spinors and so we work in the space $m=0$. This reduces the above solution to the Dirac eq:

$$\not{P} u_{\pm} = 0$$

$$\not{P} v_{\pm} = 0$$

$$(\not{\epsilon} \gamma^{\mu})^T = \epsilon_0 \not{\epsilon} \gamma^{\mu}$$

$$\gamma^{\mu} \gamma^{\nu} + \gamma^{\nu} \gamma^{\mu} = -2\eta^{\mu\nu} \mathbb{1}$$

Define the action of $(\gamma^{\mu})^{\dagger}$

$$(\gamma^{\mu})^T \epsilon^{\dagger} = \epsilon_0 \not{\epsilon} \gamma^{\mu}$$

$$\otimes (\gamma^{\mu})^{\dagger} \epsilon^{\dagger} = \epsilon_0 \not{\epsilon}^* (\gamma^{\mu})^*$$

$$(\gamma^{\mu})^* = \epsilon_0 \not{\epsilon}^* (\gamma^{\mu})^* (\epsilon^{\dagger})^{-1}$$

* Her. properties come from the invariance of $\mathcal{L}$:

$$\mathcal{L}^{\dagger} = \mathcal{L}$$

$$(i \psi^{\dagger} \gamma^0 \gamma^{\mu} \partial_{\mu} \psi)^{\dagger}$$

$$= -i \partial_{\mu} \psi^{\dagger} (\gamma^{\mu})^{\dagger} (\gamma^0)^{\dagger} \psi$$

$$\gamma^0 \gamma^0 + \gamma^0 \gamma^0 = -2(-1) \mathbb{1}$$

$$(\gamma^0)^2 = 1 \Rightarrow$$

$$i\psi^\dagger \gamma^0 (\gamma^0 \gamma^\mu (\gamma^0)^\dagger) \partial_\mu \psi \\ \equiv i\bar{\psi} \gamma^\mu \partial_\mu \psi$$

$$\gamma^\mu = \gamma^0 (\gamma^\mu)^\dagger (\gamma^0)^\dagger \quad | \text{ Since } (\gamma^0)^\dagger = \gamma^0 \\ \gamma^0 \gamma^\mu (\gamma^0)^\dagger = (\gamma^\mu)^\dagger$$

$$\boxed{(\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0} \quad \text{or}$$

$$\text{Thus: } \not{p} u_\pm = 0 \Rightarrow (\not{p} u)^\dagger = 0 \\ \partial_\mu \bar{u} (\gamma^\mu)^\dagger = 0 \\ = \partial_\mu \bar{u} \gamma^0 \gamma^\mu \gamma^0 = 0 \\ = \partial_\mu \bar{u} \gamma^\mu = 0$$

$$\text{Thus: } \bar{u}_\pm \not{p} \gamma^\mu = 0 \Rightarrow \bar{u}_\pm(p) \not{p} = 0$$

$$\boxed{\begin{aligned} * \bar{u}_\pm(p) \not{p} &= 0 \\ \not{p} u_\pm(p) &= 0 \end{aligned}}$$

* Conventions:

$(v_\pm(p), \bar{u}_\pm(p))$: outgoing
 $(u_\pm(p), \bar{v}_\pm(p))$: incoming

$v_\pm$: Anti-fermion outgoing
 $\bar{u}_\pm$: fermion outgoing

* In the massless case we associate the helicity basis $h = \pm 1/2$

Crossing symmetry:

$\left\{ \begin{array}{l} \text{(incoming} \xrightarrow{\text{yellow}} \text{outgoing)} \\ \text{(fermions} \xrightarrow{\text{red}} \text{Antifermions)} \end{array} \right. \oplus \begin{pmatrix} \text{helicity} \\ \text{flip} \end{pmatrix}$

$$v_{\pm}(p) \xrightarrow{\text{yellow}} \bar{v}_{\pm} \xrightarrow{\text{red}} u_{\mp}$$

$$\therefore u_{\pm} = v'_{\mp}, \quad \bar{v}_{\pm} = u_{\mp}$$